

Geometric structure and foundations of functional theories

Christian Schilling

in collaboration with:

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+ many further students and collaborators



Why density (matrix) functional theory?

⊕ increasing system size $\Rightarrow$ wave function methods unfeasible

⊕ highly economical:  $\xRightarrow[\text{effort}]{\text{joint}}$ universal functional

wave function
methods

versus

functional
theories

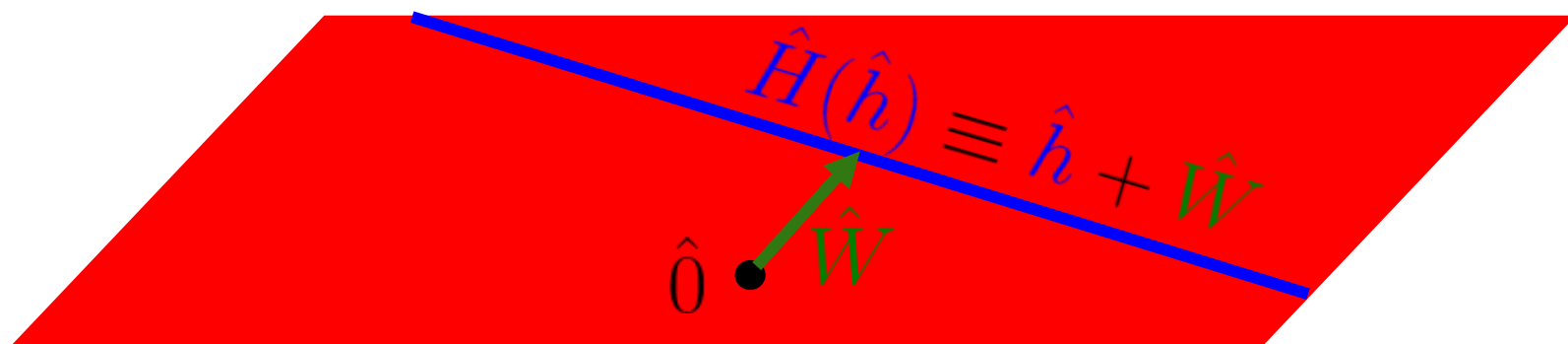
most economic!

$$E = \min_{|\Psi_N\rangle} \langle \Psi_N | \hat{H} | \Psi_N \rangle$$

$\forall \hat{H} \rightarrow$ overkill !?

interaction $\hat{W}$ fixed

$$E_{\hat{W}}(\hat{h}) = \min_{\hat{\gamma}} \left[\underbrace{\langle \hat{h}, \hat{\gamma} \rangle}_{\text{conjugate variables}} + \mathcal{F}_{\hat{W}}(\hat{\gamma}) \right]$$



all Hermitian operators on $\mathcal{H}_N$
exponentially large!

constrained search

$$\begin{aligned} E(\hat{h}) &= \min_{|\Psi_N\rangle} \langle \Psi_N | \hat{h} + \hat{W} | \Psi_N \rangle \\ &\quad \swarrow \text{1-particle Hamiltonian} \\ &= \min_{\hat{\gamma}} \min_{|\Psi_N\rangle \mapsto \hat{\gamma}} \left[\text{Tr}_1[\hat{h}\hat{\gamma}] + \langle \Psi_N | \hat{W} | \Psi_N \rangle \right] \\ &\equiv \min_{\hat{\gamma}} \left[\text{Tr}_1[\hat{h}\hat{\gamma}] + \mathcal{F}(\hat{\gamma}) \right] \rightarrow \boxed{\text{RDMFT}} \end{aligned}$$

transformative insight

in general:

variational principle + scope $\hat{H}(\hat{h}) \equiv \hat{h} + \hat{W}$

$\Rightarrow$ functional theory

- variable
- functional

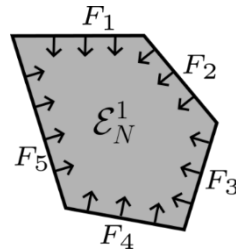
[J.Liebert, A.Y.Chaou, **CS**, J. Chem. Phys. 158, 214108 (2023)]

[J.Liebert, M.Penz, **CS**, C.-C.Wang, forthcoming]

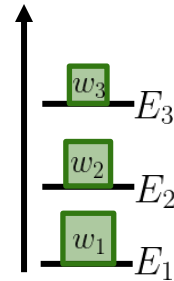
Outline

1) N-representability & convex relaxation

2) Boundary forces

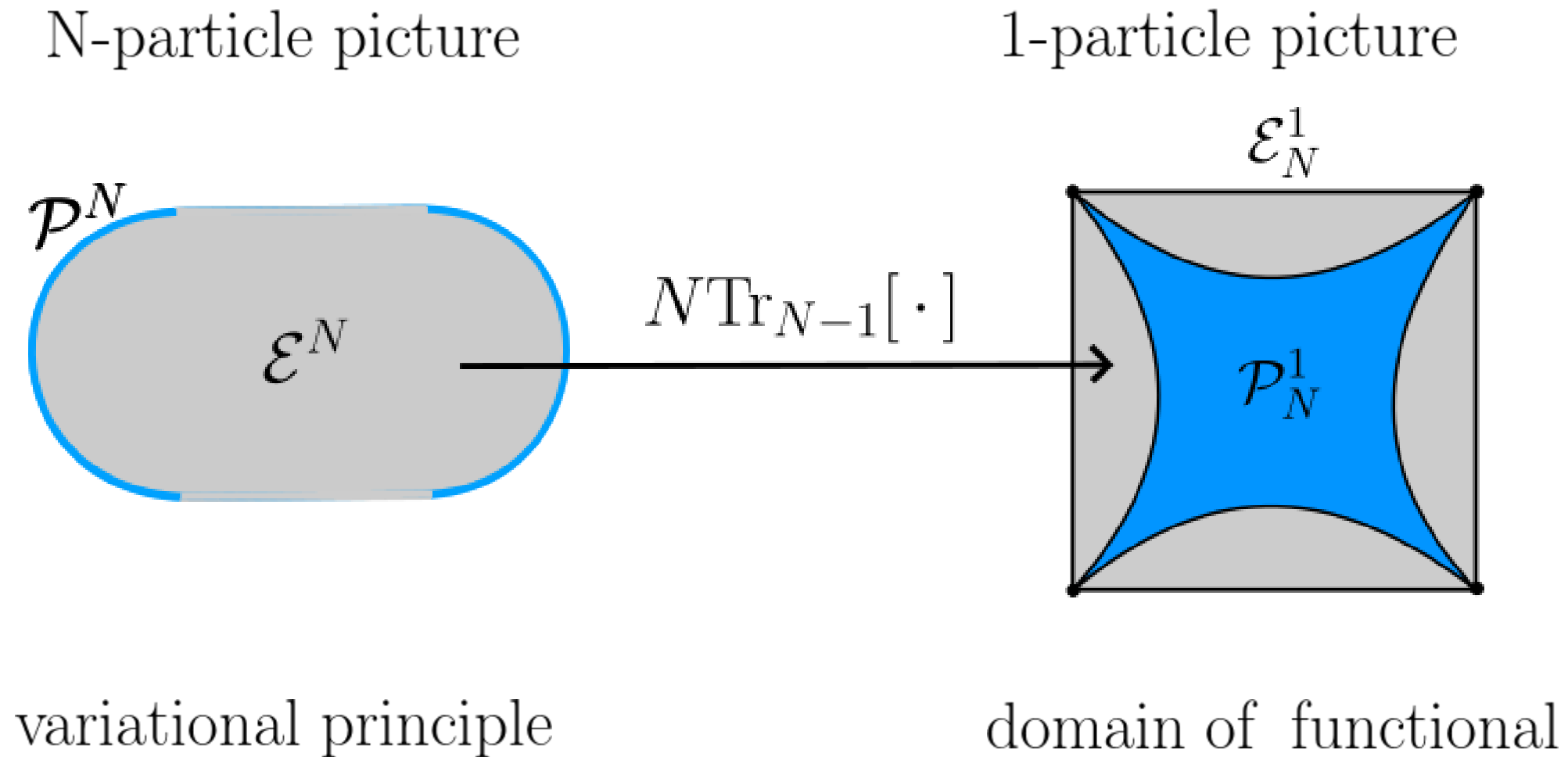


3) w-ensemble RDMFT



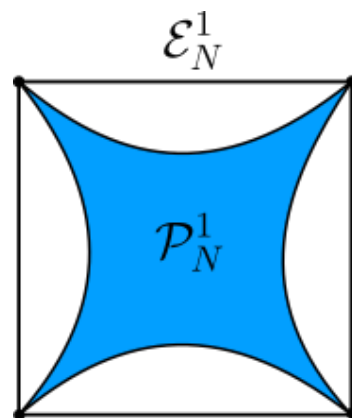
(I) N-representability & convex relaxation

Geometric picture of quantum states



pure RDMFT

$\text{Tr}_1[h\gamma] + \mathcal{F}_p(\gamma)$
minimized on $\mathcal{P}_N^1$



ensemble RDMFT

$\text{Tr}_1[h\gamma] + \mathcal{F}_e(\gamma)$
minimized on $\mathcal{E}_N^1$

ground state energy $E(h)$
ground state 1RDM γ

[T.L.Gilbert, Phys. Rev. B **12**, 2111 (1975)]

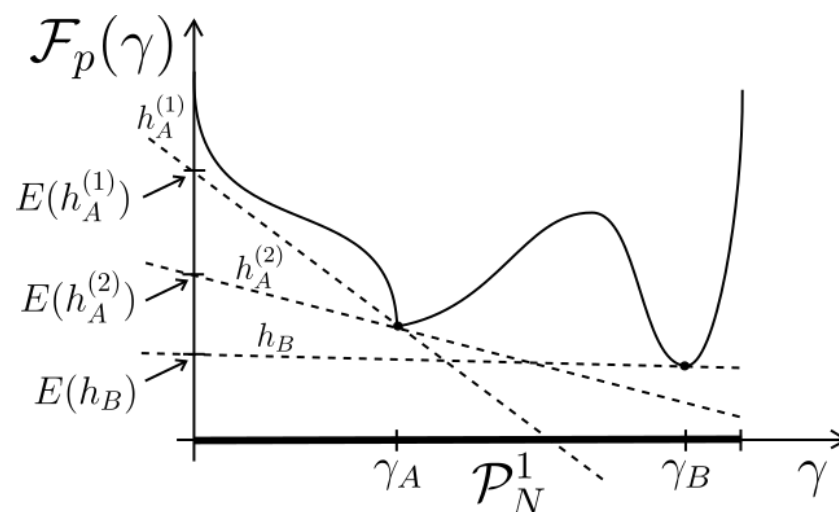
[M.Levy, PNAS **76**, 6062 (1979)]

[S.M.Valone, J. Chem. Phys. **73**, 1344 (1980)]

Relation between $\mathcal{F}_p$ and $\mathcal{F}_e$

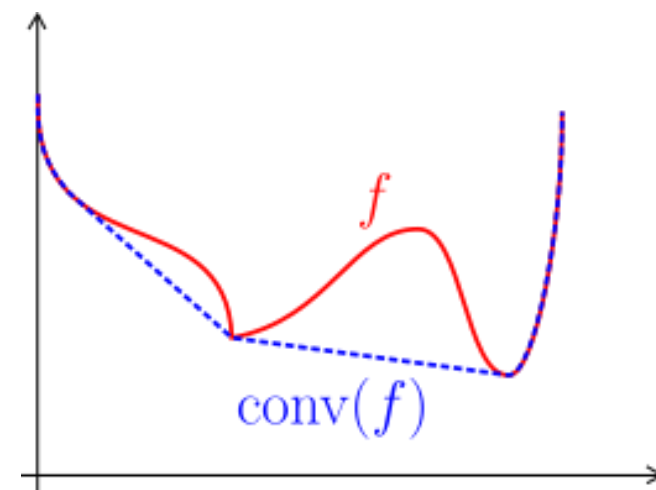
$$E(h) = \min_{\gamma} \left[\text{Tr}_1[h\gamma] + \mathcal{F}_p(\gamma) \right] \equiv \mathcal{F}_p^*(h)$$

conjugate/Fenchel-Legendre transform



crucial theorem in Convex Analysis:

$$f^{**} = \text{conv}(f)$$



since $E = \mathcal{F}_p^* = \mathcal{F}_e^*$:

$$\left. \begin{array}{l} E^* = \mathcal{F}_e^{**} = \text{conv}(\mathcal{F}_e) = \mathcal{F}_e \\ E^* = \mathcal{F}_p^{**} = \text{conv}(\mathcal{F}_p) \end{array} \right\} \Rightarrow \boxed{\mathcal{F}_e = \text{conv}(\mathcal{F}_p)}$$

$\nwarrow$
 $\mathcal{F}_e$ convex

(II) Boundary forces

Domain of 1RDM functional

fermions:

$$\langle \Psi | f_{\varphi}^{\dagger} f_{\varphi} | \Psi \rangle \leq 1$$

+generalized Pauli constraints

[CS, J. Chem. Phys. 149, 231102 (2018)]

bosons:

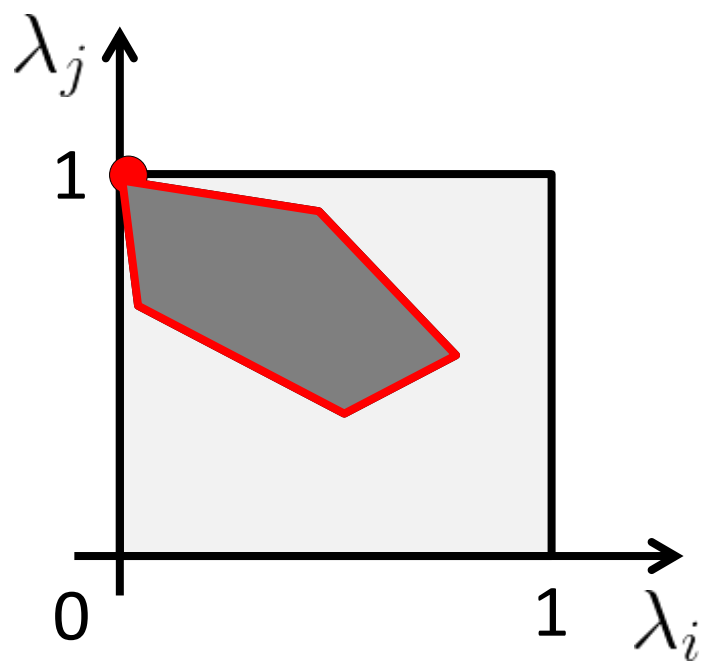
$$\langle \Psi | b_{\varphi}^{\dagger} b_{\varphi} | \Psi \rangle \leq N$$

[C.Benavides, J.Wolff, M.Marques, **CS**,
Phys. Rev. Lett. 124, 180603 (2020)]

fundamental result in  QIT:

extremal $\rho_1 \Rightarrow$ simpler structure of $|\Psi_N\rangle$

Implication of (quasi)pinning



$$\vec{\lambda} = (\underbrace{1, \dots, 1}_N, 0, \dots) \Rightarrow |\Psi_N\rangle = |1, \dots, N\rangle$$

stable

$$\vec{\lambda} \approx (\underbrace{1, \dots, 1}_N, 0, \dots) \Rightarrow |\Psi_N\rangle \approx |1, \dots, N\rangle$$

Theorem (in words):

$$\vec{\lambda} \in \partial\mathcal{P} \text{ (pinning)} \Rightarrow \text{simplified structure of } |\Psi_N\rangle$$

Theorem:

$$D(\vec{\lambda}) \equiv \kappa_0 + \sum_{j=1}^d \kappa_j \lambda_j = 0 \quad \Rightarrow \quad \hat{D}_{\Psi_N} |\Psi_N\rangle \equiv D(\{\hat{n}_i\}) |\Psi_N\rangle = 0$$

refers to natural orbitals of $|\Psi_N\rangle$

Superselection rule for Slater determinants:

- $|\Psi_N\rangle = \sum_{1 \leq i_1 < \dots < i_N \leq d} c_{i_1, \dots, i_N} |i_1, \dots, i_N\rangle$
- $|i_1, \dots, i_N\rangle$ are the eigenstates of $\hat{D}_{\Psi_N}$

check whether $\hat{D}_{\Psi_N} |i_1, \dots, i_N\rangle = 0$

if ‘no’ $\stackrel{\text{Theorem}}{\Rightarrow} c_{i_1, \dots, i_N} = 0$

Stability of structural implications:

Theorem

$$1 - \|\hat{P}_D \Psi\|^2 \leq 2D(\vec{\lambda})$$

projector onto subspace of pinning

[CS, C.Benavides-Riveros, P.Vrana, Phys. Rev. A 96, 052312 (2017)]



$$1 \longleftrightarrow N$$

[A.Klyachko, arXiv:0904.2009]

[CS, Phys. Rev. A 91, 022105 (2015)]

[CS, C.Benavides, P.Vrana, Phys. Rev. A 96, 052312 (2017)]

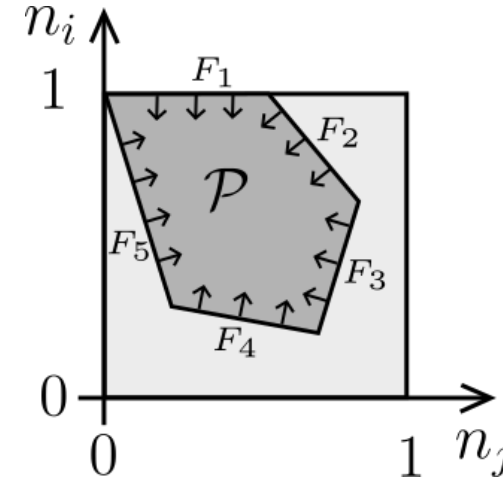
[CS, C.L.Benavides-Riveros, A.Lopes, T.Maciazek, A.Sawicki,
New J. Phys. 22, 023001 (2020)]

[T.Maciazek, A.Sawicki, D.Gross, A.Lopes, CS, New J. Phys. 22 023002 (2020)]



Theorem/conjecture:

$$\frac{\partial \mathcal{F}}{\partial \rho_1} \propto \frac{1}{\sqrt{\text{dist}(\rho_1, \partial P)}}$$



fermions

PHYSICAL REVIEW LETTERS **122**, 013001 (2019)

Diverging Exchange Force and Form of the Exact Density Matrix Functional

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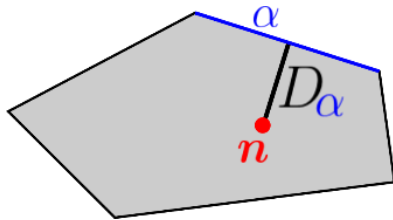
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 (Received 28 August 2018; published 4 January 2019)

‘exchange force’

$\Rightarrow n_i = 0, 1$: impossible!

$\Rightarrow$ generalizes Fermi pressure



$$\mathcal{F}(\mathbf{n}) = \sum_{\alpha, \beta} W_{\alpha\beta} \sqrt{D_{\alpha}(\mathbf{n}) D_{\beta}(\mathbf{n})}$$

bosons

PHYSICAL REVIEW LETTERS **124**, 180603 (2020)

Reduced Density Matrix Functional Theory for Bosons

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 (Received 16 February 2020; accepted 21 April 2020; published 7 May 2020)

‘BEC force’

$\Rightarrow$ complete BEC impossible

$\Rightarrow$ origin of quantum depletion

+ [J. Liebert, **CS**, Phys. Rev. Research 3, 013282 (2021)]

+ [T. Maciazek, New J. Phys. 23, 113006 (2021)]

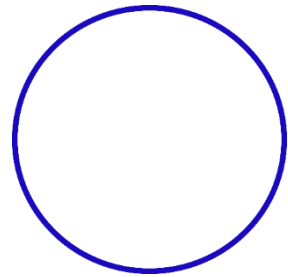
+ [J. Liebert, **CS**, SciPost Phys. 14, 120 (2023)]

$$\mathcal{F}(\mathbf{n}) = \sum_p W_p \left[n_p - \sqrt{n_p(n_p + 1)} \right]$$

+ [C.-C. Wang, **CS**, arXiv:2601.01652]

(III) w-ensemble RDMFT

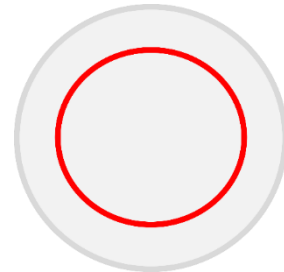
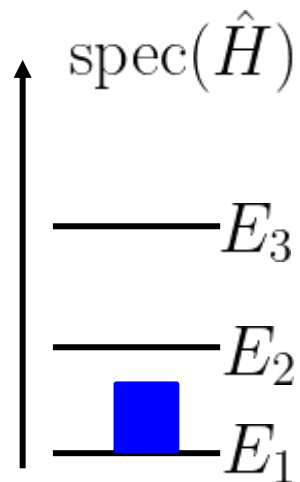
variational principle $\rightarrow$ minimizing $\text{Tr}[\hat{H}\hat{\Gamma}]$



$$|\Psi_1\rangle\langle\Psi_1|$$



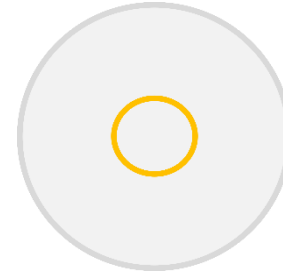
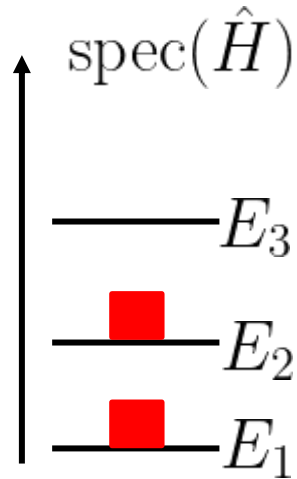
$$E_1$$



$$\frac{1}{2} [|\Psi_1\rangle\langle\Psi_1| + |\Psi_2\rangle\langle\Psi_2|]$$



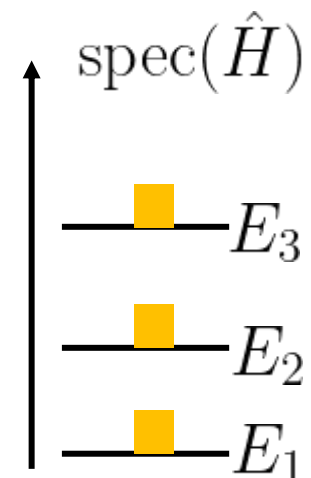
$$\frac{1}{2} [E_1 + E_2]$$



$$\frac{1}{3} [|\Psi_1\rangle\langle\Psi_1| + |\Psi_2\rangle\langle\Psi_2| + |\Psi_3\rangle\langle\Psi_3|]$$



$$\frac{1}{3} [E_1 + E_2 + E_3]$$

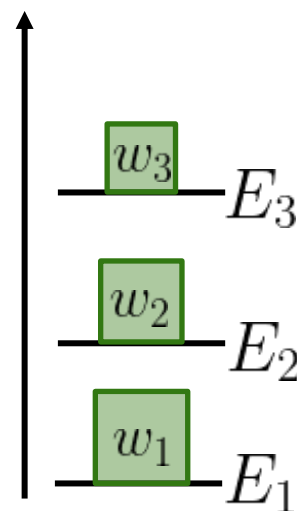


GOK variational principle

$$E_{\mathbf{w}} \equiv \sum w_j E_j = \min_{\hat{\Gamma} \in \mathcal{E}^N(\mathbf{w})} \text{Tr}[\hat{H}\hat{\Gamma}] \quad \leftarrow \text{spec}(\hat{\Gamma}) = \mathbf{w}$$

[E.K.U.Gross, L.N.Oliveira, W.Kohn, Phys. Rev. A 37, 2805 (1988)]

[L.Ding, C.-L.Hong, **CS**, Quantum 8, 1525 (2024)]



↓ + constrained search

$\mathbf{w}$ -ensemble RDMFT

{ universal functional $\mathcal{F}_{\mathbf{w}}(\hat{\gamma})$
domain: $\mathcal{E}_N^1(\mathbf{w}) = N \text{Tr}_{N-1}[\mathcal{E}^N(\mathbf{w})]$

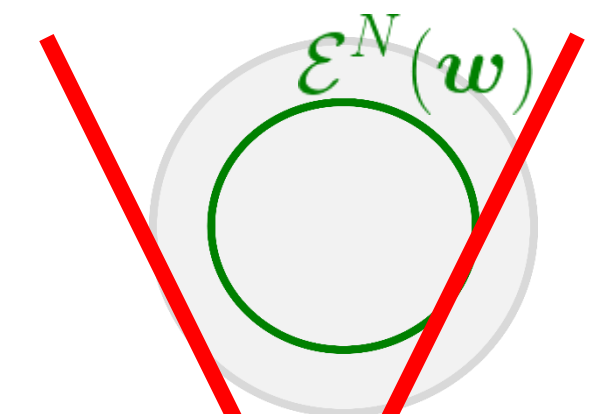
[**CS**, S.Pittalis, Phys. Rev. Lett. 127, 023001 (2021)]

[J.Liebert, F.Castillo, J.-P.Labbé, **CS**, J. Chem. Theory Comput. 18, 124 (2022)]

[J.Liebert, **CS**, New J. Phys. 25, 013009 (2023)]

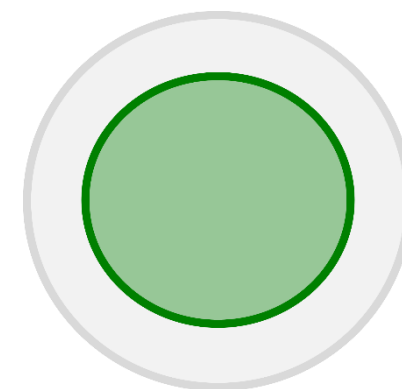
domain of $\mathcal{F}_w(\hat{\gamma})$?

N



key idea

extend
space

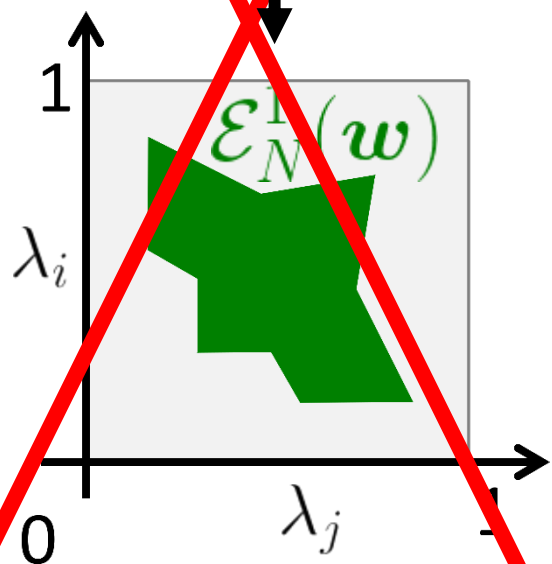


Theorem

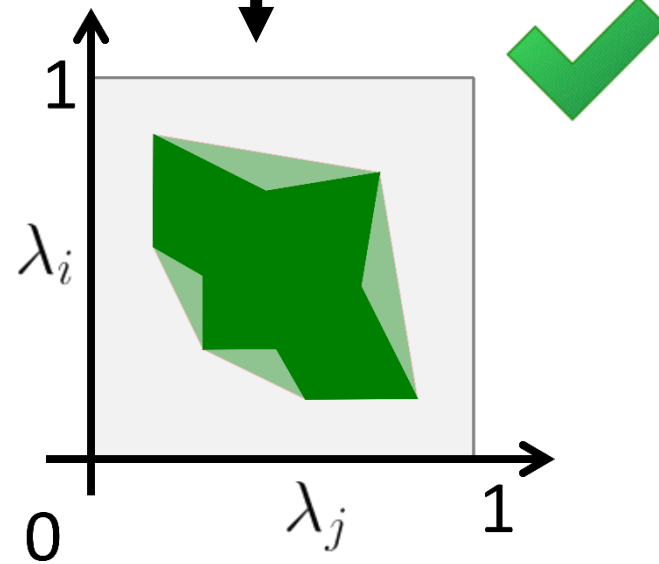


convex
relaxation

1



too complicated

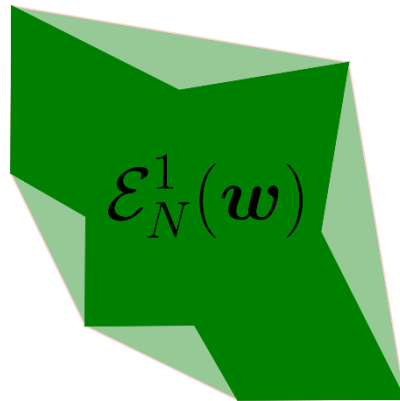


[CS, S.Pittalis, Phys. Rev. Lett. 127, 023001 (2021)]

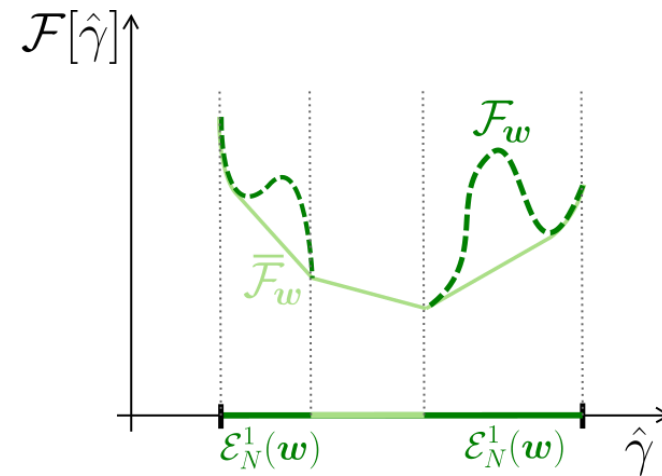
[J.Liebert, F.Castillo, J.-P.Labbé, CS, J. Chem. Theory Comput. 18, 124 (2022)]

exact convex relaxation

domain



functional



“lower convex envelope”

hierarchy of exclusion principles

setting

■ $\boldsymbol{w} = (w_1, 0, \dots)$

■ $\boldsymbol{w} = (w_1, w_2, 0, \dots)$

■ $\boldsymbol{w} = (w_1, w_2, w_3, 0, \dots)$

constraints

$$\lambda_1^\downarrow \leq 1$$

+

$$\sum_{j=1}^N \lambda_j^\downarrow \leq N - 1 + w_1$$

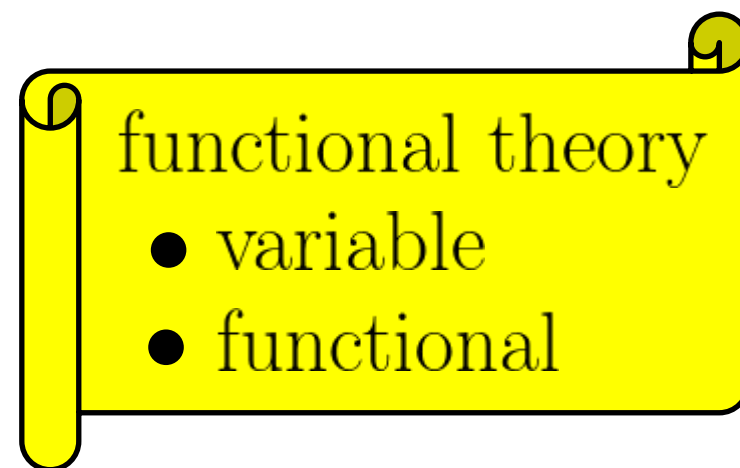
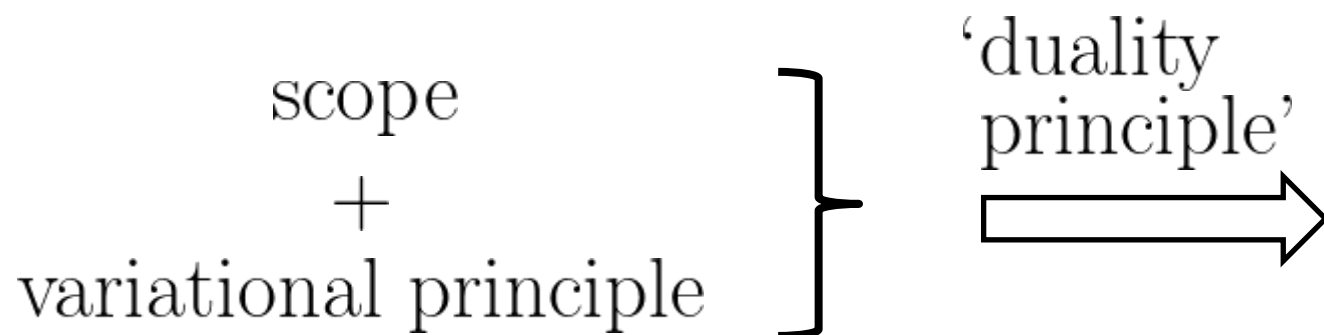
+

$$2 \sum_{j=1}^{N-1} \lambda_j^\downarrow + \lambda_N^\downarrow + \lambda_{N+1}^\downarrow \leq 2N - 2 + w_1 + w_2$$

Summary



transformative insight:



geometry of density matrices + Convex Analysis:

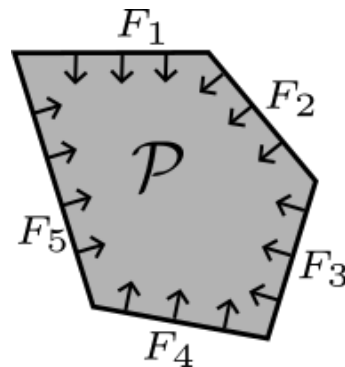
$$\mathcal{F}_e = \text{conv}(\mathcal{F}_p)$$

⇒ complexity of GPCs transferred into functional?

Summary

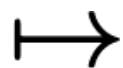


Dynamical interpretation of
(generalized) Pauli constraints
also for: bosons, spins, etc.
analogously: (lattice) DFT



w -ensemble variational principle + constrained search
 $\Rightarrow$ w -RDMFT for excitation energies
 $\Rightarrow$ complete hierarchy of exclusion principles

Schilling group



UNIVERSITÉ
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Thank you!



Postdoc/PhD positions

e.g., *BeyondDFT*



Conso-
lidator
Grant



<https://sites.google.com/view/christianschilling>